Details

- **Course: IFT 211**
- **Lab Assessment: Smart Water Tank Monitoring & Control System**
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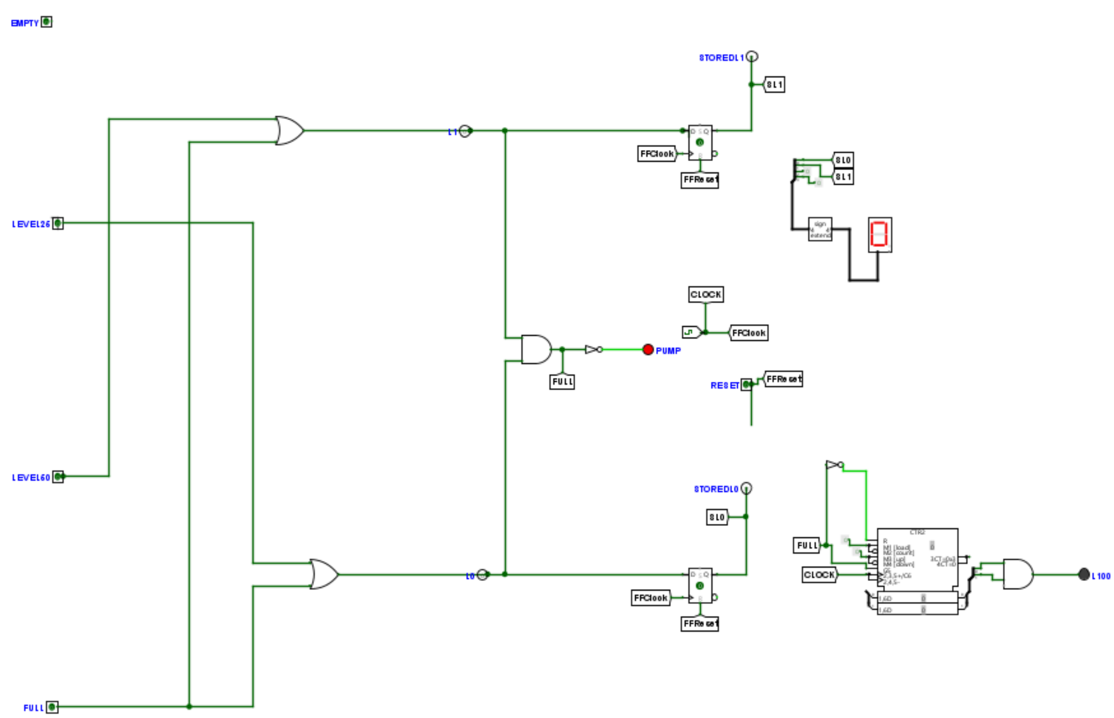
- **Matric No: 2024/B/CSC/0960**
- **Date:07/13/2026**

1.Logisim circuit file : Circuit Design (Logism) Below!



2. Boolean Derivations:

Pump:

Pump = ¬Full

Alarm:

Alarm = Q1 · Q0

Register:

StoredL1 = L1 (captured on FFClock)

StoredL0 = L0 (captured on FFClock)

3. Truth Tables:

Pump Truth Table (Pump = ¬Full)

L1	L0	FULL	PUMP
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Alarm Truth Table (Alarm = Q1 · Q0)

Q1	Q0	ALARM
0	0	0
0	1	0
1	0	0
1	1	1

Register Truth Table (Register stores last valid tank level)

L1	L0	StoredL1	StoredL0
0	0	0	0
0	1	0	1
1	0	1	0
1	1	1	1

4. K-map simplifications:

Pump K-Map

Pump = ¬Full
Variables: L1, L0

	L0=0	L0=1
L1=0	1	1
L1=1	1	0

Pump = 1 everywhere except when L1=1 and L0=1 (Full).

Alarm K-Map

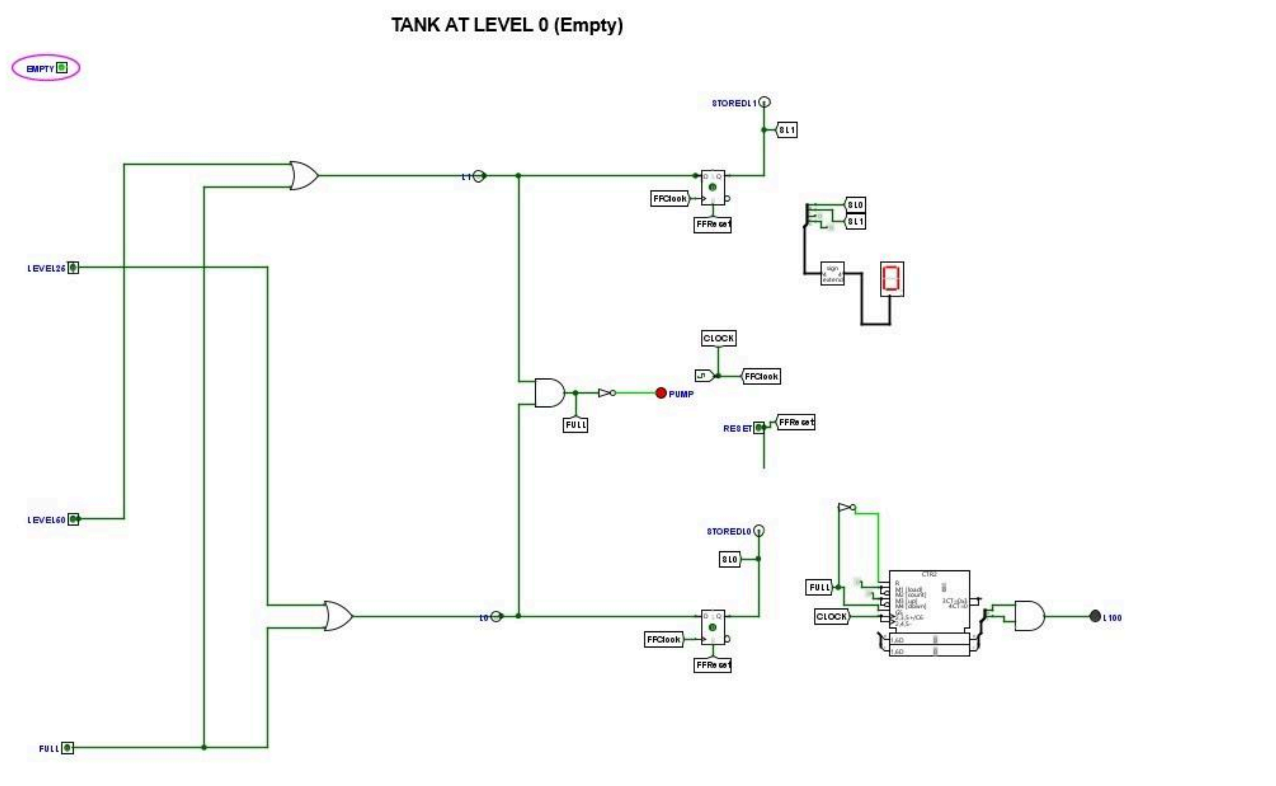
Alarm = Q1 · Q0
Variables: Q1, Q0

	Q0=0	Q0=1
Q1=0	0	0
Q1=1	0	1

Alarm = 1 only when Q1=1 and Q0=1.

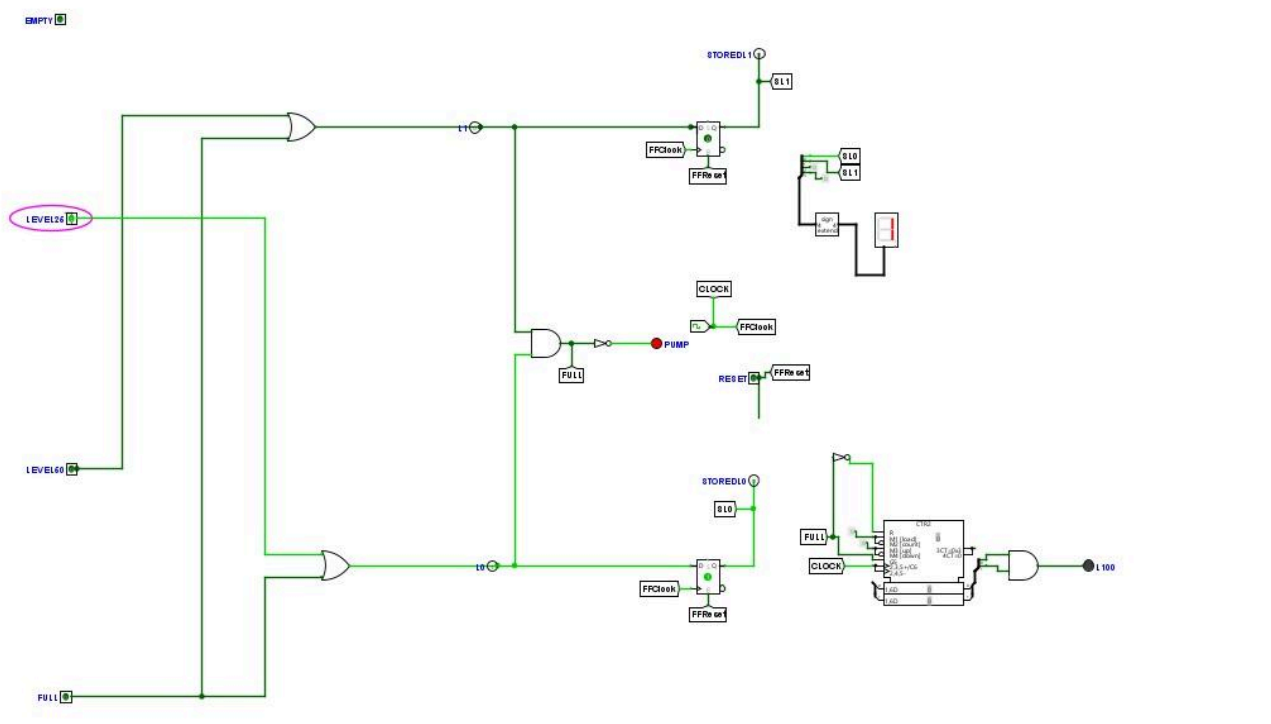
5.Testing Scenarios:

- **Empty (0%) → Pump ON, Alarm OFF**

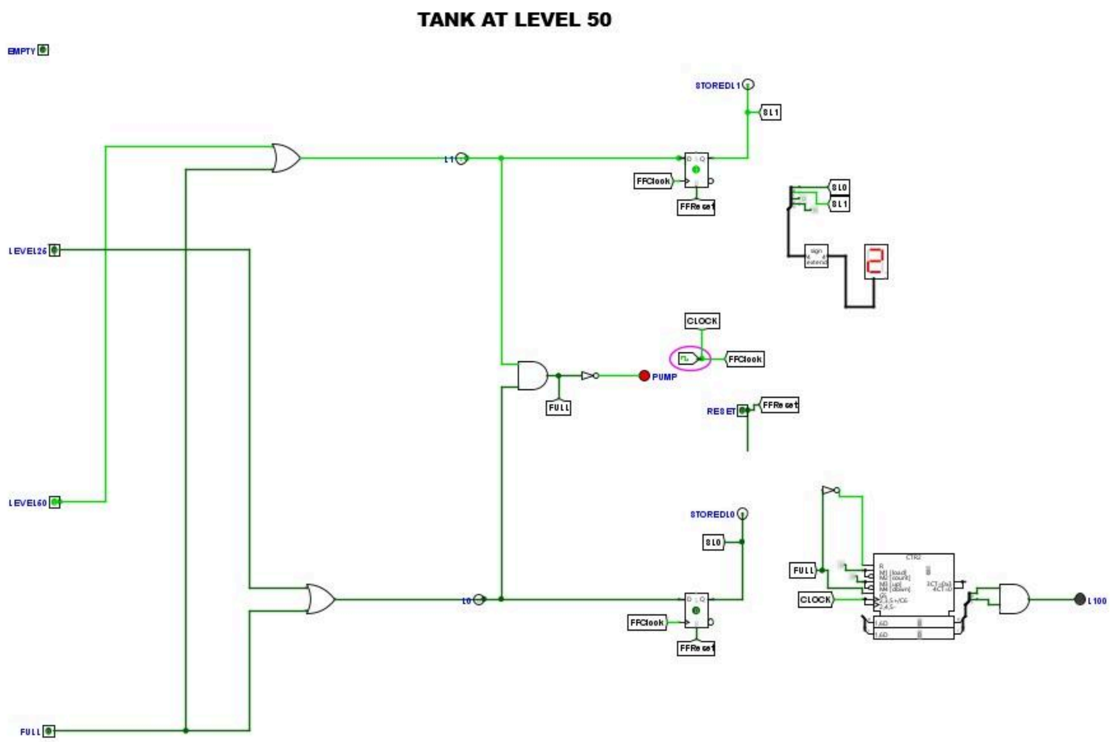


● **25% Level → Pump ON, Alarm OFF**

TANK AT LEVEL 25

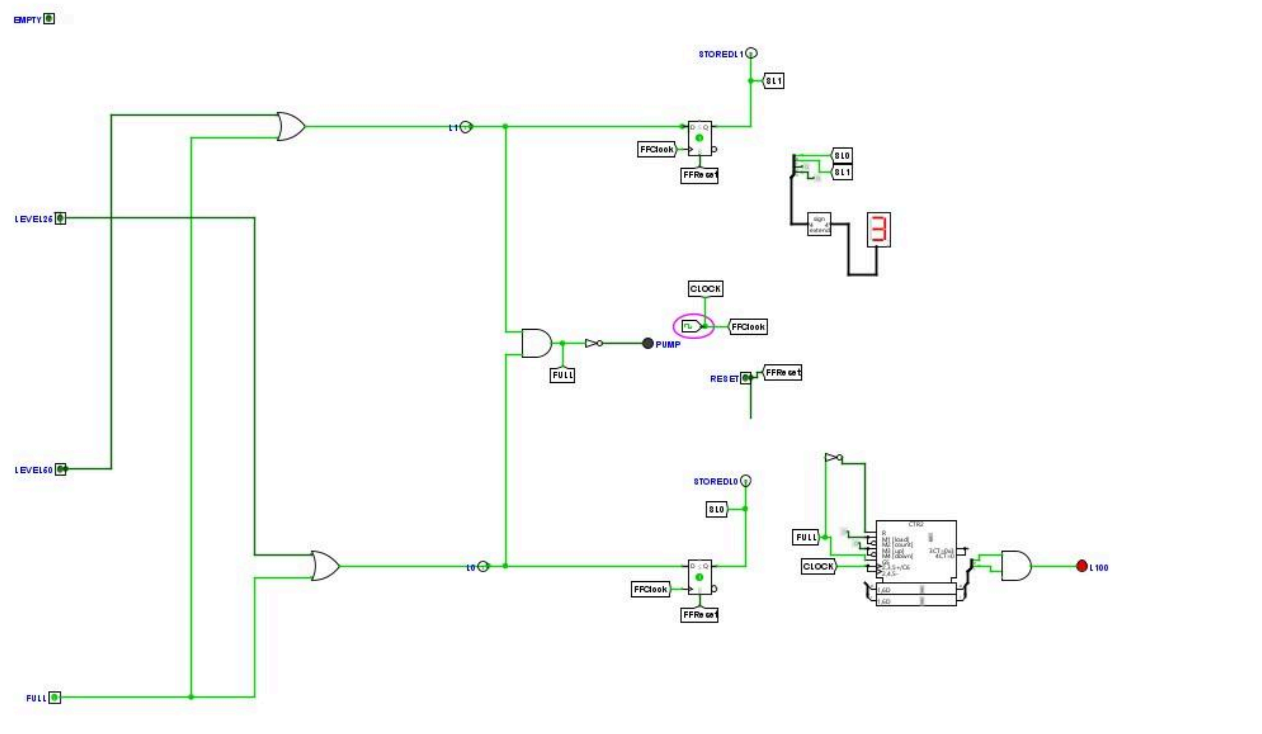


- **50% Level → Pump ON, Alarm OFF**



- **Full (100%) → Pump OFF, Alarm ON (after 3 cycles of sustained FULL condition**

TANK AT LEVEL 100 (FULL)



6. System Explanation:

The Smart Water Tank Monitoring and Control System is designed to regulate water levels automatically and provide safety alerts when abnormal conditions occur. The system uses four sensors to detect the tank's water level (Empty, 25%, 50%, and Full).

- **Pump Operation: The pump is controlled directly by the sensor outputs. It turns ON whenever the tank is not full and turns OFF immediately when the tank reaches the full level. This ensures the tank is always filled to a safe level without overflowing.**
- **Alarm Operation: The alarm is controlled by a 2-bit counter connected to the Full signal. When the tank reaches full, the counter begins counting clock cycles. If the Full condition persists for three consecutive cycles, the counter output reaches 11 and activates the alarm. This prevents false alarms and ensures the alarm only signals a sustained overflow condition.**
- **System Behavior: Together, the pump and alarm provide a reliable monitoring system. The pump reacts instantly to water level changes, while the alarm only activates under prolonged overflow. This separation of functions ensures efficient water management and safety.**